

RESEARCH ARTICLE

Customer-Oriented Boundary Spanning, Functional Diversity, and Customer Adoption

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ABSTRACT

For teams developing products or services to meet customer needs, customer adoption of team deliverables is core to their success. When such teams focus on complex, tailored deliverables, customer adoption can be expected to benefit from team information elaboration—the exchange, discussion, and integration of team members' knowledge and perspectives—to develop solutions for customer needs. We aim to shed light on how teams can focus on the customer's perspective within the elaboration process to drive customer adoption. We propose that whereas engaging with the customer's perspective is key to customer adoption, teams may only do this to a modest degree unless they are stimulated to put the customer perspective center-stage. Extending information elaboration theory by drawing on the attention-based view, we propose that customer-oriented boundary spanning—engaging with the customer to champion the customer's perspective within the team—strengthens the shared objective of serving the customer to guide information elaboration and increase the quality of knowledge work. We argue that this effect is moderated by team functional background diversity: increased attention to the customer's perspective guides teams to better use their informational resources and this benefit is stronger with greater functional background diversity. These predictions were supported in a field experiment with a customer-oriented boundary spanning intervention ($N = 144$ teams). Shared objectives and information elaboration sequentially mediated the effect of customer-oriented boundary spanning and the indirect effect from customer-oriented boundary spanning to customer adoption was stronger with greater functional diversity.

For teams developing products or services to meet customer needs, customer adoption of team deliverables is key to team and firm success (Heidenreich et al. 2024). This makes customer adoption a major performance indicator for professional service teams (Anand et al. 2007; Cromwell and Gardner 2020). However, research shows that firms struggle in achieving customer adoption (Hoda et al. 2011; Karp 2023). This raises the question, what influences team effectiveness in developing solutions that result in customer adoption? Core to answering that question is to recognize that professional service teams engage in knowledge work delivering tailored solutions to customers. In knowledge work, teams can be seen as information processing systems for which team information elaboration—the exchange,

discussion, and integration of task-relevant information and perspectives (van Knippenberg et al. 2004)—is the key process driving the quality of their performance (Hinsz et al. 1997; De Dreu et al. 2008).

Accordingly, for customer adoption, the more teams connect with their customer's perspective in information elaboration, the more teams may expect adoption of the solutions they develop, as these will be more informed by customer needs (Gruber et al. 2015). However, team members typically know less about the customer's perspective compared to their own perspectives (Gruber et al. 2015) and team members naturally focus on their own functional perspective in the information

elaboration process (Cronin and Weingart 2007; Edmondson and Harvey 2018; Edmondson and Reynolds 2016; Hoever et al. 2012; Micheli et al. 2019). A key challenge for these teams then is not just to stimulate team information elaboration but to direct attention to the customer's perspective in the elaboration process. This challenge is the focus of our study.

To address this challenge, we extend information elaboration theory, the perspective emphasized by the various team knowledge work research streams (De Dreu et al. 2008; Gardner et al. 2012; Resick et al. 2014; Samba et al. 2018; van Knippenberg 2017; van Knippenberg et al. 2004), with the core insight from the attention-based view that a key influence on knowledge integration is the focus of attention in information processing (Brielmaier and Friesl 2023; Haas et al. 2015; Ocasio 1997, 2011). This theoretical integration allows us to propose that knowledge team performance benefits from an influence that not only stimulates information elaboration but also focuses the team's attention on delivering on customer needs. We capture such influence through the consideration of *customer-oriented boundary spanning*.

Customer-oriented boundary spanning refers to team member interaction with the customer to acquire information and to coordinate with the customer for team task performance (Bayighomog and Araslı 2019; Bettencourt and Brown 2003; cf. Ancona and Caldwell 1992). Boundary spanning more broadly defined as connecting to others outside of the team has been recognized as a positive influence on team innovation (Alexander and van Knippenberg 2014; Ancona and Caldwell 1992; Marrone 2010; van Knippenberg et al. 2024) and on knowledge integration across team boundaries (Mell et al. 2022). Customer-oriented boundary spanning specifically has been identified as a positive influence in teams' ability to serve customers (Bettencourt and Brown 2003; Faraj and Yan 2009; Long et al. 2007). Building on this work, we argue that customer-oriented boundary spanning focuses the team's attention on the shared objective of delivering on customer needs. By doing so, customer-oriented boundary spanning stimulates information elaboration with a customer focus (cf. the attention-based view of information processing; Haas et al. 2015; Ocasio 1997), with greater customer adoption as a result.

The proposition that customer-oriented boundary spanning has a positive effect on customer adoption, while novel, is a relatively straightforward extrapolation from the literature. We advance this proposition primarily to set the stage for the main contribution of our study: the proposition that customer-oriented boundary spanning has stronger effects with greater team functional background diversity.

We argue through the attention to the customer's perspective it stimulates, customer-oriented boundary spanning leads teams to make better use of their informational resources in developing customer deliverables. As a result, the positive effect of customer-oriented boundary spanning on customer adoption is stronger with greater team informational resources. We capture these resources through team functional background diversity, a team informational resource strongly tied to team knowledge integration (Hülshager et al. 2009; van Dijk et al. 2012; van Knippenberg 2017). This focus on functional background diversity is well-aligned with a core implication of information elaboration theory that to capture what drives team performance in knowledge work, we need to look at sources of diverse perspectives as drivers of information elaboration. In delivering complex, tailored solutions to customers, teams perform better the more they achieve integration of team members' diverse knowledge and perspectives (Gardner et al. 2012; Resick et al. 2014). Indeed, the fact that teams can draw on diverse expertise and perspectives is a core reason that strategic decision-making, innovation, and professional service delivery typically is the responsibility of teams (Barczak et al. 2009; Glissmann et al. 2010; Hülshager et al. 2009; Mathieu et al. 2017; Samba et al. 2018; Thayer et al. 2018; van Knippenberg 2017). Accordingly, the combined focus on customer-oriented boundary spanning and functional diversity allows us to capture the integration of core insights from information elaboration theory (van Knippenberg 2017; van Knippenberg et al. 2004) and the attention-based view (Ocasio 1997, 2011). Figure 1 captures our research model.

Our study advances team information elaboration theory and research. By taking a multi-theory approach, we bridge information elaboration theory (van Knippenberg et al. 2004) and the attention-based view (Ocasio 2011) to recognize that team information elaboration—as the key driver of team knowledge work performance—is driven by both the informational resources available to the team and by attentional focus in information elaboration. Information elaboration theory suggests that information elaboration is driven by the motivation and ability to engage in information elaboration. As such, it implies that with greater motivation and ability, team informational resources are more fully utilized through information elaboration. While this may hold all other things being equal, the attention-based view allows us to recognize that attention determines what information is drawn on through information elaboration such that higher information elaboration may primarily reflect higher information elaboration *with a particular focus of attention*. The current focus on customer-oriented boundary spanning thus moves beyond earlier work on influences motivating information elaboration

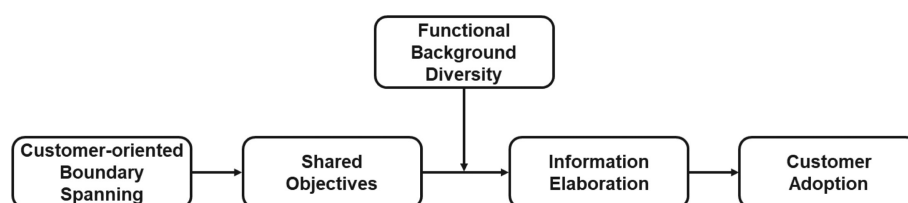


FIGURE 1 | Research model.

to capture customer-oriented boundary spanning's influence in focusing the team's attention to make better use of its informational resources in pursuit of customer adoption of the outcomes produced by the team. Our study also provides a meaningful extension of the customer success and adoption literature as it recognizes the challenges associated with aligning external customer needs with technology solution capabilities (Frambach and Schillewaert 2002; Karp 2023; Lucas et al. 2007). Furthermore, we are able to identify the interactive influence of customer-oriented boundary spanning and functional background diversity as a way to meet these challenges. We test our research model in a field experiment in which customer-oriented boundary spanning is induced through a designated boundary spanning team member, which means our insights are not only supported by causal evidence but also actionable in that organizations can adopt our boundary spanner intervention to boost customer adoption.

1 | Theoretical Background and Hypotheses

Stimulating customer adoption of a product, service, business process, program, or procedure is challenging for customer-facing teams that develop and deliver such solutions (West 2002). This is illustrated by evidence across industries, such as health-care (Chen et al. 2020), higher education (King and Boyatt 2015), and banking (Keskar and Pandey 2018). Part of the challenge here for teams that deliver tailored services and solutions to their customers in complex business settings are the knowledge work requirements to deliver solutions that are to some extent unique to a particular customer (Cromwell and Gardner 2020; Frambach and Schillewaert 2002; Glissmann et al. 2010).

For example, professional services teams, such as management consulting teams, are knowledge-intensive teams that provide tailored solutions for their customers and rely on customer adoption as a primary indicator of success (Anand et al. 2007; Gardner 2016). For such teams, a focus on the customer's perspective is key to successful performance because customers will only adopt solutions to the extent they meet their needs. Part of the challenge of meeting customer needs for teams providing tailored solutions is that to some extent each solution is unique to the customer. This differentiates such teams from teams that develop products for a broader user market. Teams that develop products to bring to market also benefit from designing with the user in mind (Cousins 2018; Elsbach and Stigliani 2018; Maruping and Matook 2020). Because of their market focus however, they benefit more from market research than from understanding one individual customer's needs (Hoda et al. 2011; Stock 2014). In contrast, for teams delivering tailored solutions to individual customers, even when they are developed from the firm's base products (e.g., such as found in complex business-to-business ICT products and services; Heidenreich et al. 2024; Desyllas et al. 2023; Mehta and Pickens 2020), a focus on the specific customer the team is developing solutions for is more beneficial. This is where customer-oriented boundary spanning has particular value in introducing a focus on the customer's perspective to the team.

For team knowledge work, outcomes follow from the integration of information and perspectives. This approach has

been articulated in team diversity research, where the concept of team information elaboration was first proposed (van Knippenberg et al. 2004) as well as team innovation research (van Knippenberg 2017). Its roots exist in the notion of teams as "information processing systems" (Hinsz et al. 1997; De Dreu et al. 2008) such that, for any kind of team knowledge work, information elaboration is the core process driving team performance. We refer to this broader perspective on team knowledge work as information elaboration theory (cf. Resick et al. 2014). In the present study, we extend information elaboration theory by integrating insights from the attention-based view (Ocasio 1997, 2011) to suggest that knowledge work benefits from a shared focus of attention in information elaboration, stimulated by customer-oriented boundary spanning.

Team information elaboration research has by and large seen antecedents of information elaboration as concerning information processing motivation and ability, and team informational resources (De Dreu et al. 2008; van Knippenberg et al. 2004). Team informational resources (e.g., team diversity) reflects what is available to the team for information elaboration (an issue we revisit when we discuss the role of functional background diversity). Team member motivation and ability reflects the extent to which team members are motivated and able to draw on team informational resources through team information elaboration. In the present study, we draw on both motivation and ability as well as informational resources to argue for the influence of customer-oriented boundary spanning in motivating focused information elaboration. We suggest this allows teams to make better use of their functional background diversity. We further extend this argument with the attention-based view (Ocasio 1997, 2011) to position influences on attention in elaboration as a specific form of motivation.

For instance, team information elaboration research has shown that teams tend to engage in more information elaboration the more team members have psychological traits reflecting a motivation for open-minded information processing, such as openness to experience (Homan et al. 2008), need for cognition (Kearney et al. 2009), and learning goal orientation (Nederveen Pieterse et al. 2013). Team member cognitive ability likewise has been found to predict information elaboration (Resick et al. 2014). Motivation for information elaboration can also follow from an understanding that such elaboration can facilitate team performance benefits. For example, research by van Ginkel, van Knippenberg and colleagues (van Ginkel and van Knippenberg 2008, 2009, 2012; Kooij-De Bode et al. 2008) demonstrates that understanding the value of information elaboration to team performance motivates elaboration, as do factors that help team members to develop such an understanding (cf. Mell et al. 2014). In a related vein, Resick et al. (2014) found that cognitive ability influences information elaboration through an understanding of the team's task. Our current focus on customer-oriented boundary spanning as motivating information elaboration by focusing attention on the customer's perspective is consistent with this broader emphasis on motivation in information elaboration theory. As such, we complement and extend this theory by recognizing that attention in information elaboration is a motivational expression. Specifically, attention reflects which team informational resources members are motivated to draw on as they engage in informational elaboration.

1.1 | Customer-Oriented Boundary Spanning, Information Elaboration, and Customer Adoption

Boundary spanning refers to interactions outside of unit boundaries to acquire information, to coordinate efforts between units, and to garner support for a unit's pursuit of its objectives (Faraj and Yan 2009; Marrone et al. 2007; Porck and van Knippenberg 2023; Schotter et al. 2017). Broadly defined, boundary spanning is a positive influence on team performance (Ancona and Caldwell 1992; Howell and Shea 2006; Hülshager et al. 2009; Somech and Khalaili 2014), which is well-aligned with information elaboration theory and research (Mell et al. 2022; van Knippenberg and Hoefer 2021; van Knippenberg et al. 2024). Following the recognition that key to customer adoption is an emphasis on the customer's perspective within the team, our focus is more specifically on customer-oriented boundary spanning—boundary spanning that specifically concerns customer contacts. Customer-oriented boundary spanning activities include ambassadorial activities to support customer-focused goals, access and mobilize critical resources as well as scouting activities to gather relevant customer information, and task-coordinator activities to orchestrate key team processes focusing on the customer. Of core relevance to our information elaboration perspective, customer-oriented boundary spanning is a way to introduce external information to the team that specifically concerns the customer's perspective as well as to encourage the team to focus information elaboration around the customer's perspective.

The attention-based view on knowledge integration in organizations captures why such attention-focusing influence is important (Brielmaier and Friesl 2023; Haas et al. 2015; Ocasio 1997, 2011). As Simon (1957) observed in his seminal work, the scarce factor in organizations is not information but attention. Knowledge workers are faced with more information than they can process. It is the focus of their attention that determines which information is processed (Ocasio 1997, 2011). Information elaboration theory and the attention-based view converge on the notion that while informational resources are important as input for team information elaboration, having the resource does not equate to using the resource. What the attention-based view suggests explicitly however, is only implied in information elaboration theory (cf. De Dreu et al. 2008): The motivation and ability to elaborate on team informational resources does not guarantee that all resources receive equal attention in information elaboration; attention influences which information is primarily drawn on. Accordingly, an influence focusing attention in information elaboration on a specific issue—such as customer-oriented boundary spanning—is critical to ensuring the team focuses on the most pertinent information driving team success. We propose that customer-oriented boundary spanning not only introduces information about the customer to the team but also serves to focus attention on the customer's perspective by stimulating the shared objective of delivering on customer needs.

Shared objectives and goals at work that provide the basis for action within the team (Anderson and West 1998), are a positive influence in team knowledge work performance (Hülshager et al. 2009). A focus on shared objectives provides a coordination point for team members that allows the team to make better use of

its (human) resources (Fay et al. 2006; Lyubovnikova et al. 2018; Pearce and Ensley 2004; van Knippenberg et al. 2011). We propose that by stimulating the shared objective of addressing the customer's needs, customer-oriented boundary spanning stimulates information elaboration. Importantly, this elaboration will not just concern the client information that customer-oriented boundary spanning may introduce. By providing a clear reference point in the shared objective to address the customer's perspective, customer-oriented boundary spanning makes it easier for team members to more broadly recognize which knowledge, information, and perspectives to share, discuss, and integrate in their teamwork for the customer. Customer-oriented boundary spanning thus stimulates the team to make more, and more focused, use of its informational resources through information elaboration. This leads to our first hypothesis:

Hypothesis 1a. *Shared objectives mediate the positive effect of customer-oriented boundary spanning on information elaboration.*

By stimulating information elaboration guided by the shared objective to serve the customer, customer-oriented boundary spanning can be expected to result in greater customer adoption. Indeed, teams that engage in higher levels of information elaboration yield better performance outcomes (van Ginkel and van Knippenberg 2008). In our context, the higher and more focused information elaboration engendered by customer-oriented boundary spanning can be expected to stimulate the development of customer deliverables that are better aligned with customer needs. Because customers can be expected to show greater adoption of deliverables that better match their needs (Gruber et al. 2015), customer-oriented boundary spanning can be expected to have a positive effect on customer adoption that is sequentially mediated by shared objectives and information elaboration. This is not to deny the value of diversity of perspectives for knowledge work or that similarity in perspectives may be associated with lower creativity (Desyllas et al. 2023; Skilton and Dooley 2010). Rather, it is to recognize that shared objectives—a shared understanding of what the team is focused on—allows the team to make better use of its diversity of information, insights, and perspectives (cf. Acar et al. 2019, on the creativity and innovation benefits of focus, and Samba et al. 2018, on the decision-making benefits of shared objectives). Accordingly, we advance the following hypothesis.

Hypothesis 1b. *Shared objectives and information elaboration sequentially mediate the positive effect of customer-oriented boundary spanning on customer adoption.*

1.2 | The Moderating Role of Functional Background Diversity

Developing our analysis further, we propose that the influence of customer-oriented boundary spanning in stimulating shared objectives, information elaboration, and customer adoption implies that the effect of customer-oriented boundary spanning is stronger for teams with greater functional background diversity. The more proximal influence involved in the influence of customer-oriented boundary spanning in stimulating focused elaboration exists in the shared objectives

it stimulates. Accordingly, moderation by functional background is understood to occur in the relationship between shared objectives and information elaboration. It is shared objectives that channel teams attention (cf. Ocasio 1997) to make more focused use of its informational resources—and greater functional diversity means greater informational resources. Thus, in considering the interplay of customer-oriented boundary spanning and functional background diversity, we can more fully flesh out the implications of integrating insights from the attention-based view into information elaboration theory.

To drive adoption success (especially for complex products and services in a business to business context), a variety of expertise and experience is required (e.g., process design, change management, user training, and solution customization; cf. Heidenreich et al. 2024; Karp 2023). Functional background diversity reflects task-relevant differences in knowledge, information, and perspectives. Through information elaboration, teams with greater functional background diversity can be more creative and innovative. This has been established meta-analytically (Hülsheger et al. 2009; van Dijk et al. 2012) as well as in primary research (Hoever et al. 2012; Hoever et al. 2018; Moser et al. 2019; Somech and Khalaili 2014; Zhang 2016; for a review, see van Knippenberg and Hoever 2021). In a similar vein, Gardner et al. (2012) found in their study of professional service teams that such teams have to systematically integrate knowledge drawing on their functional diversity to improve performance outcomes. Furthermore, team decision making studies identify information elaboration as the key process through which teams use their distributed information and expertise to make higher-quality decisions (Mell et al. 2014; van Ginkel and van Knippenberg 2008).

Recognizing that functional background diversity represents a valuable informational resource, we propose the elaboration-stimulating effect of customer-oriented boundary spanning through shared objectives is stronger for teams with greater functional background diversity. As per the rationale for Hypothesis 1a, customer-oriented boundary spanning stimulates and focuses team information elaboration by directing team members attention to the customer's perspective. A core element in this logic is that customer-oriented boundary spanning, by stimulating shared objectives, guides the team to make better use of its informational resources. The boundary to this effect implied by this rationale is that it should hold to the extent the team has informational resources to elaborate. Specifically, the team can only exchange, discuss, and integrate the diversity of knowledge, information, and perspectives members bring to the team. Accordingly, the shared objective to serve the customer that follows from customer-oriented boundary spanning can stimulate information elaboration to stronger effect with greater functional background diversity (i.e., functional background diversity moderates the path from shared objectives to information elaboration). Conversely, with less functional background diversity, the effect of customer-oriented boundary spanning on information elaboration will be weaker.

Hypothesis 2a. *The positive effect of customer-oriented boundary spanning on information elaboration through shared*

objectives is stronger (weaker) with greater (smaller) functional background diversity.

As a direct consequence of this interactive effect on information elaboration, customer-oriented boundary spanning will have a stronger effect on customer adoption for teams with greater functional background diversity. This is because customer-oriented boundary spanning stimulates information elaboration focused on customer needs and thus channels the team to better—from a customer adoption perspective—use its informational resources. The more informational resources the team has—the greater its functional diversity—the more team performance in terms of customer adoption benefits from this focused elaboration. We also recognize information overload is a reality of knowledge work, and by drawing from the attention-based view (Ocasio 2011), we need not only consider the team's informational resources as input for information elaboration but also factors influencing the focus of attention in team information elaboration. Attention to information is key as—attention determines what is and is not integrated in teamwork outcomes. This translates to how team members allocate their collective attention. Accordingly, we predict second-stage moderation by functional background diversity in the effect of customer-oriented boundary spanning on customer adoption, such that this effect, sequentially mediated by shared objectives and information elaboration, is stronger with greater (and weaker with smaller) functional background diversity.

Hypothesis 2b. *The indirect effect of customer-oriented boundary spanning on customer adoption, sequentially mediated by shared objectives and information elaboration, is stronger (weaker) with greater (smaller) functional background diversity.*

2 | Methods

2.1 | Research Design

Our research design combines four different data sources and a field experimental intervention, using a pretest-posttest experimental group—control group design to test the effect of customer-oriented boundary spanning on team information elaboration and customer adoption. The field experiment was conducted over a 6-month period and involved 162 customer engagement teams of an IT service provider working actively with North American customers across industries. These customer engagement teams consist of people with different functional expertise sharing the responsibility for team-level tasks to explore new business opportunities and apply enterprise software and technology solutions by continuously engaging with their customers (cf. Kozlowski and Bell 2003). Exploration includes problem identification focused on customer needs, ideation of solutions that address the needs and problems, and eventually the implementation of those solutions to address the identified customer needs (March 1991).

The teams in our study delivered customized IT solutions to external clients. This typically requires creativity in generating solutions that are aligned with customer needs. Many firms

therefore directly work with customers to develop solutions specifically for them (e.g., Heidenreich et al. 2024; Desyllas et al. 2023; Mehta and Pickens 2020). The business interest is that the customers display high usage (implementation) of the solution because this is expected to result in a continued relationship. Accordingly, the company that hosted our research put significant emphasis on customer usage of its customized IT solutions as indicator of success. In this context, boundary-spanning can help to collaboratively develop a mutual understanding and address all aspects of the business relationship, including the strategy, business operations, solution functionality, organization and partner alignment, risk mitigation, and support services (Edmondson and Harvey 2018; Hoda et al. 2011; Maruping and Matook 2020; Stock 2014). Thus, there is a clear case that for value realization in firms like this, customer-oriented boundary spanning can be an important positive influence in ensuring that teams develop solutions that are implemented by their clients.

To assess the effect of customer-oriented boundary spanning in a way that allowed conclusions about causality, we conducted a field experiment that revolved around the introduction of a designated customer-oriented boundary spanning role into teams in our treatment condition (to be compared with teams in our control condition). Teams in the treatment condition were assigned a team member that was, as per the definition of customer-oriented boundary spanning, charged with the role of bridging the team and the customer to bring in information about the external client and focus the discussion to stimulate the processing of that information. This role complemented existing sales and marketing roles with the post-sales tasks of coordinating resources across departments (and implementation partners), focusing on the customer's business outcomes, ongoing process governance (e.g., deployment planning, feature adoption, user training, guidance for optimal solution usage, support, and ongoing system health monitoring), and communication. Specifically, in communication with the customer the boundary-spanning team members helped to mitigate the risk that it is used more to communicate what the team is doing than to get input from the customer. Team boundary spanners also directed team attention to input from the customer and prioritize accordingly. In other words, the overall orchestration of the implementation and adoption process by understanding the fundamental problems to be solved, ideating the best solution, and creating alignment for team goals. The boundary-spanning team members were trained for these specific activities before introducing them to the team and their customers to ensure that all boundary spanners operated under the same set of guidelines and operating parameters. It is also noteworthy that the company did a pilot test of the boundary-spanning intervention with 30 customers, which resulted in several refinements of the engagement methodology and the training materials that were later utilized in the field experiment. Upon the beginning of their engagement, the boundary spanners were tasked to organize a customer relationship assessment and with it, set expectations and objectives, and align strategy no matter in which engagement phase the customer was at the time. To assess customer adoption, we relied on objective customer usage data measured at the beginning (i.e., before the treatment) and the end of the experiment, thus creating a pretest-posttest

experimental group—control group design. Notably, teams in the control group were not assigned a customer-oriented boundary spanner and yet were tasked with the same objectives as the experimental condition.

Information about team information elaboration along with control variables, such as team tenure, work experience, and executive support was collected via survey, and we utilized company archival data to gather information on functional background diversity and a time-lagged objective measure of customer adoption.

2.2 | Data and Sample

The unit of analysis is the customer engagement team with team sizes ranging from 3 to 27 and a median of 9 members ($SD = 4.36$), including but not limited to sales reps, industry and value advisors, solution and support engineers, enterprise architects, engagement managers, as well as product and adoption experts. In the experiment, teams were randomly assigned to either the treatment or control group. In 48% of the teams with treatment status, an existing team member was assigned to the new role and their old position was backfilled (i.e., team-internal assignment for the boundary spanning role). In 20% of the cases, boundary spanners served multiple customer engagement teams in his or her role. These attributes represent arrangements in the treatment condition (i.e., internal assignment and multi-team service), which are deviations from the ideal design with a designated customer-oriented boundary spanning role in which all treatment teams would be the same on these attributes, but these deviations were inevitable in the study context where the intervention had to be aligned with business realities. Importantly however, supplementary analysis showed that neither internal assignment ($\beta = -0.18$, $SE = 0.15$, $p = 0.255$) nor multi-team service ($\beta = 0.19$, $SE = 0.16$, $p = 0.227$) had a significant effect on customer adoption.¹

Our survey instrument was piloted individually with five subject matter experts. Based on the feedback, we made some minor wording changes such that participants better understand that the questions reference the team environment and related mechanisms. We sought responses from the company employees who had been part of the experiment for the full 6 months in either the treatment or control group.

A total of 935 team members completed the survey, yielding a response rate of 62%. Across all team members, 75% were male and 25% female. 54% of the survey respondents reported a bachelor's and 41% a master's degree; the remaining 5% were high school and doctorate degrees. Average work experience of participants was 23.5 years ($SD = 9.31$). Thirty-nine responses (3%) did not specify their team and were not included in the analysis. Individual team member responses were aggregated to the team level by computing the means resulting in 162 team cases. Sixteen teams with only one response and two treatment teams where the customer did not consent to participation in the study were not included in the analysis. The remaining $N = 144$ teams are the basis for the analysis, of which 77 received the treatment, and 67 constituted the

control group. These 144 teams did meet the sampling ratio cut-off value of 0.17 (Dawson 2003; Richter et al. 2006) and had response rates of at least 60%.

2.3 | Measures

2.3.1 | Customer-Oriented Boundary Spanning

Customer-oriented boundary spanning was dummy-coded to differentiate the control (0) and treatment (1) groups.

2.3.2 | Shared Objectives

In alignment with previous research (e.g., van Knippenberg et al. 2011), we used six items from Anderson and West (1998) to measure shared objectives.² For example, “How clear are you about what your team’s objectives are?” and “To what extent do you think other team members agree with these objectives?” Team members were asked to rate each item for their team on a 7-point scale ranging from “not at all” to “completely.” Cronbach’s alpha of 0.95 indicated excellent internal homogeneity and reliability. The calculated agreement index median R_{wg} of 0.87 and intraclass correlations $ICC_{(1)}$ of 0.74 and $ICC_{(k)}$ of 0.95 justified aggregation to the team level (LeBreton and Senter 2008).

2.3.3 | Functional Background Diversity

Based on our theoretical approach and in alignment with previous research (e.g., Kukenberger and D’Innocenzo 2020; Zhang 2016), we operationalized the diversity of team member functional backgrounds with Blau’s index of heterogeneity normalized for varying team sizes as recommended by Biemann and Kearney (2010) across the following 11 categories of business functions: Distribution & Logistics, Finance & Accounting, Human Resources, Information Technology, Marketing & Communications, Production, Research & Development, Supply Chain & Operations, Sales & Customer Service, Sourcing & Purchasing, and Strategy & Business Development. We calculated the diversity index for each team based on available survey responses and company records for team members that did not submit their survey. The index ranges from zero indicating no functional background diversity to one indicating maximum diversity of functional backgrounds among team members. The functional diversity measure includes the background of the boundary-spanning team member. The most common functional backgrounds were sales and customer services (26%) and information technology (20%). Seventeen percent of the team members described their function as “other” and provided detailed backgrounds, such as supply chain, sales and distribution, logistics, and production in the aerospace industry.

2.3.4 | Information Elaboration

To assess information elaboration, we used the 6-item scale of van Knippenberg, van Ginkel, Stam, Giessner, van Balen, and Ghorbani (2019; cf. van Knippenberg et al. 2024). Sample items include “In our team members complement each other by sharing

their knowledge” and “We discuss alternatives before making decisions.” As part of the survey, respondents were asked to indicate their agreement to these items within their team on a 5-point scale ranging from “strongly disagree” to “strongly agree.” Cronbach’s alpha of 0.96 indicated excellent internal reliability. We calculated R_{wg} and intraclass correlations (ICCs) to assess the appropriateness of aggregation to the team level (LeBreton and Senter 2008). For the information elaboration measure, median R_{wg} of 0.87 and intraclass correlations $ICC_{(1)}$ of 0.80 and $ICC_{(k)}$ of 0.96 justified aggregation to the team level.

2.3.5 | Customer Adoption

Customer adoption was measured as the usage of an enterprise software technology solution within the adopting customer organization. The utilization of the implemented technology solution was represented as a percentage of the actual metered usage over the purchased software entitlement. For example, if a customer purchased 500 users of a particular solution and 100 users actually used it, the adoption index would be 20%. The utilization percentages were calculated for all participating customers with an analytics report of the company. They were made available for the beginning (baseline) and at the end of the measurement period. The index ranges from zero indicating no use to one indicating full use. In case a customer used multiple software solutions integrated into one business process, then the adoption index was calculated as the mean across all solutions. Examples include implementing recruiting and talent management solutions, replacing a legacy enterprise resource planning system, and exploring new solutions to make supply chains more resilient.

2.3.6 | Treatment Checks

In order to assess the effectiveness of the experimental intervention, we used the following direct and indirect treatment checks (Ejelöv and Luke 2020). First, we checked directly whether a boundary-spanning team member was assigned to the treatment group teams. This information was available in the data set of the company experiment. Beyond this essential check, we analyzed whether the boundary spanners performed the trained intervention. As elaborated above, boundary spanners were trained to start or reset the customer engagement with a relationship assessment workshop and document goals and progress in an outcome success plan. In the survey, we asked team members to what extent they agree or disagree (5-point scale ranging from “strongly disagree” to “strongly agree”) that an outcome success plan was developed by the team and shared with the customer since this represents the most salient and tangible measure of the intervention.

Furthermore, we used Faraj and Yan’s (2009) scale to measure boundary spanning team activities with our survey instrument. Respondents were asked to indicate their agreement to the four items on a 5-point scale ranging from “strongly disagree” to “strongly agree.” For example: “Does your account team value team members for making use of their relationships with others on behalf of the team?” Cronbach’s alpha of 0.91 indicated excellent reliability. For the boundary spanning treatment check, a

median R_{wg} of 0.79 and intraclass correlations $ICC_{(1)}$ of 0.70 and $ICC_{(k)}$ of 0.90 justified aggregation to the team level.

2.3.7 | Controls

We included executive support, team tenure, and team size, and work experience, customer familiarity and industry as control variables in the analyses to partial out their effects on customer adoption. Notably, we also controlled for adoption baseline levels which allowed us to empirically account for endogeneity. Within the treatment group we also controlled for assignment and multi-team support cases.

Team size was measured as the sum of team members. We included team size as a control variable because it is often found in team research that it influences team performance (Fay et al. 2006; Hülsheger et al. 2009; Moser et al. 2019). Research evidence suggests that smaller and larger teams differ in many important ways. While larger teams can enjoy access to more resources, higher levels of diversity, and may appear more legitimate, they can also experience more coordination and communication challenges, member task conflict, and less cohesion (Mathieu et al. 2008; Moreland, Levine, and Wingert 1996 in Witte and Davis 2014).

Executive sponsorship and top management support have been identified as critical success factors for innovation with information systems (e.g., Chandler and Thomas 2015; Lucas et al. 2007; Shao et al. 2017; Sharma and Yetton 2003), specifically because of resource intensive implementation projects and the ongoing need for end-user support and training. Executive sponsors are typically important not only during the initial stages of a customer engagement to ensure alignment and help mobilize required resources but are also critical for ongoing governance, business reviews, conflict resolution, and prioritization (Drucker 2002). Furthermore, van Meerkerk and Edelenbos (2017) found an indirect positive relationship between executive support and boundary spanning. Therefore, we also controlled for differences in executive support. We have asked team members to rate how pleased they are with the executive sponsorship of their customer engagement using a 5-point scale ranging from “*extremely displeased*” to “*extremely pleased*.”

Team tenure was also included as a control variable since it has been identified as a critical component in team effectiveness models. Gonzalez-Mulé et al. (2020), for example, found that team tenure was positively related to team performance. Zhang (2016) furthermore found that team longevity positively moderated the relationship between functional diversity and team creativity through knowledge sharing and group cohesion. Following Kozlowski et al. (1999), we calculated collective team tenure as the mean amount of time individual members have been together on the same team. Beyond the collective tenure, building on Becker's (1964 in Gonzalez-Mulé et al. 2020) human capital theory, we measured the work experience of the team, which is also referred to as additive team tenure capturing the differences in skills, experience, and knowledge available within the team. This is consistent with team composition research (e.g., Bell 2007), where individual attributes of team members are aggregated to team-level factors. Work experience

has been captured with a simple survey question asking respondents for how many years they have been in the workforce. Like Haas (2010), we calculated it as the mean amount of years that team members have spent in the workforce.

Finally, we controlled for the industry segment of the customers. The industry data for each team and customer has been obtained from company records. Based on the customer's industry they were assigned to one of the following six industry clusters: Consumer, services, manufacturing, energy and natural resources, financial services, and public sector. The industry cluster assignment for each team was coded with dummy variables and the reported results for each industry are to be interpreted in reference to public sector. Besides the industry, we also controlled for customer familiarity measured as the number of days since becoming a customer.

3 | Results

Table 1 reports the descriptive statistics and correlations for all study variables.

Customer adoption was positively related to functional background diversity and customer-oriented boundary spanning, and highly correlated with the adoption baseline (which is not surprising given that it is a repeated measure of the same variable). We also note that boundary spanning is correlated with functional background diversity despite random assignment to conditions. The boundary spanner as a team member is included in the functional background diversity measure and to determine whether this caused the correlation, we also computed the correlation without the boundary spanning member included in the diversity correlation. In this analysis, boundary spanning and functional diversity are still correlated, $r=0.32$, $p<0.001$, suggesting that it is a chance correlation and not caused by the experimental intervention. Moreover, because both are predictors in our analysis, we test the relationships for the one controlling for the other. Next, we analyzed the treatment check to establish that the treatment group engaged in more boundary spanning than the control group. The treatment group ($M=4.05$, $SD=0.40$) indeed engaged in significantly more boundary spanning than the control group ($M=3.78$, $SD=0.60$), $t(142)=3.21$, $p<0.001$.

To test the hypothesized model of relationships, we used the PROCESS macro (Hayes 2018) for SPSS Version 25. PROCESS uses an ordinary least squares regression-based path analysis to estimate direct and indirect effects in mediator models, moderation models, and conditional effects in moderated mediation models. In particular, we used the PROCESS because it allows for robust testing of relationships using bootstrapping analyses of moderated effects. We used PROCESS Model 4 for Hypothesis 1a, 1b to test the mediation effects, PROCESS model 14 for H2a to test the conditional indirect effect, and a custom model variation using PROCESS model 91 to report the index of moderated serial mediation.

The moderated mediation test is based on an interval estimate of the parameter that links the indirect effect to moderator values, providing moderated mediation indices (Hayes 2018). We used

TABLE 1 | Means, standard deviations, and correlations.

Variable	Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11
1. Customer adoption	50.08	26.78											
2. Information elaboration	4.16	0.57	0.12										
3. Functional background diversity	0.79	0.17	0.22**	0.51**									
4. Customer-oriented boundary spanning	0.53	0.50	0.21**	0.38**	0.37**								
5. Shared objectives	5.56	0.66	−0.01	0.29**	−0.04	0.18*							
6. Adoption baseline	44.53	27.25	0.78**	−0.20*	−0.08	−0.02	−0.15						
7. Executive support	3.56	0.65	−0.15	0.04	−0.08	0.14	0.39**	−0.18*					
8. Team tenure	1.78	0.41	0.06	0.16	0.10	−0.18*	0.17*	0.08	0.04				
9. Team size	8.89	4.36	−0.04	0.10	0.26**	0.30**	−0.04	0.00	−0.01	−0.02			
10. Work experience	23.11	5.39	0.07	0.25**	0.29**	0.14	0.01	−0.05	−0.05	−0.01	0.10		
11. Customer familiarity	6198	2699	−0.11	0.10	0.25**	−0.40	−0.16	−0.08	−0.12	0.14	0.22**	0.00	
12. Manipulation check	3.93	0.52	0.02	0.55**	0.23**	0.26**	0.30**	−0.15	0.16	0.12	0.17	0.12	−0.02

Note: $N = 144$.

* $p < 0.05$.

** $p < 0.01$.

the setting of 20000 iterations for the bootstrapping analyses. Bootstrapping is considered the right approach for studies of indirect effects with small sample sizes (MacKinnon et al. 2004). Table 2 reports the effects of customer-oriented boundary spanning on shared objectives, information elaboration, and customer adoption.

We first regressed shared objectives on customer-oriented boundary spanning (treatment) and the control variables, which yielded a significant R^2 of 0.29. Then we regressed information elaboration on customer-oriented boundary spanning, shared objectives, functional background diversity, their interaction terms, and the control variables, which yielded a significant R^2 of 0.63. Our results show that customer-oriented boundary spanning had a significant positive effect on shared objectives and information elaboration. Next, customer adoption was regressed on customer-oriented boundary spanning, shared objectives, functional background diversity, and their interaction terms as well as on information elaboration, the adoption baseline, and the control variables, which yielded a significant R^2 of 0.77. Finally, we tested our hypotheses by bootstrapping indirect and conditional indirect effects of customer-oriented boundary spanning—both the main effect and the effect conditional on functional background diversity as mediated by shared objectives and information elaboration. Table 3 reports the indirect

and conditional indirect effects as well as overall results of our hypotheses testing.

As reported in Table 3, the confidence interval for the indirect effect of customer-oriented boundary spanning on information elaboration, $\beta = 0.11$, $SE = 0.08$ (0.006, 0.292), excluded zero, showing that the effect was mediated by shared objectives. Therefore, Hypothesis 1a was supported. Similarly, shared objectives and information elaboration sequentially mediated the effect of customer-oriented boundary spanning on customer adoption, $\beta = 0.11$, $SE = 0.04$ (0.039, 0.208), providing support for Hypothesis 1b. Notably, we also found an indirect main effect of functional background diversity on customer adoption mediated by information elaboration, $\beta = 0.09$, $SE = 0.03$ (0.04, 0.14) that is consistent with the state of the science (van Knippenberg and Hoefer 2021). Furthermore, we found a significant first-stage moderation effect of functional background diversity on the relationship between customer-oriented boundary spanning and shared objectives ($\beta = 0.62$, $SE = 0.20$, $p < 0.001$). The interaction of shared objectives and functional background diversity displayed in Figure 2 was also significant ($\beta = 0.37$, $SE = 0.05$, $p < 0.001$). When functional background diversity was relatively high (1 SD above the mean), the effect of shared objectives on information elaboration was significant, $\beta = 0.83$, $SE = 0.10$, $p < 0.001$, whereas the effect of shared objectives was

TABLE 2 | Effects of customer-oriented boundary spanning on customer adoption, information elaboration, and shared objectives.

	Shared objectives	Information elaboration		Customer adoption	
		Model 1	Model 2	Model 1	Model 2
Constant	0.04 (0.46)	0.04 (0.42)	0.18 (0.34)	−0.50 (0.27)	−0.45 (0.27)
Customer-oriented boundary spanning	0.38 (0.16)*	0.48 (0.16)**	0.36 (0.13)**	0.23 (0.11)*	0.19 (0.11)
Shared objectives	—	0.25 (0.08)**	0.45 (0.07)**	0.05 (0.05)	0.04 (0.07)
Functional background diversity	—	0.38 (0.08)**	0.13 (0.07)	0.22 (0.06)**	0.14 (0.07)*
Functional background diversity×shared objectives	—	—	0.37 (0.05)**	—	0.04 (0.05)
Functional background diversity×customer-oriented boundary spanning	—	—	—	—	0.28 (0.13)*
Information elaboration	—	—	—	0.14 (0.06)*	0.14 (0.07)*
Adoption baseline	−0.10 (0.08)	−0.17 (0.07)*	−0.10 (0.06)	0.84 (0.05)**	0.85 (0.05)**
Team size	−0.07 (0.08)	−0.09 (0.07)	−0.03 (0.06)	−0.13 (0.05)**	−0.12 (0.05)*
Team tenure	0.25 (0.08)**	0.13 (0.07)	0.08 (0.06)	−0.03 (0.05)	−0.02 (0.05)
Executive support	0.34 (0.08)**	−0.10 (0.08)	−0.10 (0.06)	−0.01 (0.05)	−0.02 (0.05)
Work experience	−0.01 (0.08)	0.07 (0.07)	−0.04 (0.06)	0.03 (0.05)	0.03 (0.05)
Customer familiarity	−0.17 (0.08)*	0.02 (0.07)	−0.36 (0.35)	−0.07 (0.05)	−0.07 (0.05)
Consumer industries	0.01 (0.48)	−0.22 (0.44)	−0.25 (0.34)	0.38 (0.28)	0.30 (0.28)
Services industries	−0.59 (0.52)	−0.15 (0.47)	−0.38 (0.35)	0.38 (0.31)	0.28 (0.30)
Manufacturing industries	−0.31 (0.48)	−0.39 (0.43)	−0.43 (0.35)	0.40 (0.28)	0.33 (0.28)
Energy and resources industries	−0.20 (0.50)	−0.49 (0.45)	−0.43 (0.36)	0.45 (0.29)	0.38 (0.29)
Financial industries	−0.72 (0.53)	−0.06 (0.48)	−0.11 (0.40)	0.10 (0.32)	0.06 (0.31)
R^2	0.29**	0.44**	0.63**	0.76**	0.77**
F	4.48**	7.21**	14.30**	27.28**	25.02**

Note: $N = 144$. Values are standardized coefficients (with standard errors).

* $p < 0.05$.

** $p < 0.01$.

not significant, $\beta = 0.08$, $SE = 0.07$, $p = 0.247$, when functional background diversity was relatively low (1 SD below the mean).

As reported in Table 3, the confidence interval for the index of moderated mediation provides evidence that the effect of customer-oriented boundary spanning on information elaboration, $\beta = 0.14$, $SE = 0.06$ (0.017, 0.266), was mediated by shared objectives and moderated by functional background diversity. Therefore, Hypothesis 2a was supported. When functional background diversity was relatively high (1 SD above the mean), the conditional indirect effect of customer-oriented boundary spanning on information elaboration was significant, $\beta = 0.31$, $SE = 0.14$ (0.038, 0.604), whereas the effect was not significant, $\beta = 0.03$, $SE = 0.04$ (−0.038, 0.138), when functional background diversity was relatively low (1 SD below the mean).

Finally, as reported in Table 3, the positive confidence interval for the index of moderated serial mediation shows that the effect of customer-oriented boundary spanning on customer adoption, $\beta = 0.02$, $SE = 0.01$ (0.001, 0.048), was mediated by

shared objectives and information elaboration, and moderated by functional background diversity. Therefore, Hypothesis 2b was supported. The implication of our moderated mediation model is there would also be a main effect of customer-oriented boundary spanning and functional background diversity interaction on customer adoption. Our results indicate that functional background diversity also enhances the direct effect of customer-oriented boundary spanning on customer adoption, $\beta = 0.28$, $SE = 0.13$, $p = 0.032$. As illustrated in Figure 3, the pattern of this interaction was in line with predictions showing a positive effect of customer-oriented boundary spanning on customer adoption with higher functional background diversity.

3.1 | Supplementary Analysis

In 20% of the cases, boundary spanners served multiple customer engagement teams with treatment status. Our supplementary analysis for the treatment group showed that the multi-team

TABLE 3 | Indirect and conditional indirect effects of customer-oriented boundary spanning.

Indirect effect of customer-oriented boundary spanning on information elaboration (H1a)		
	Effect (SE)	95% CI
Shared objectives	0.11 (0.08)	0.006, 0.292
Conditional effects of shared objectives on information elaboration		
	Effect (SE)	95% CI
Low functional diversity (−1 SD)	0.08 (0.07)	−0.055, 0.214
Mean functional diversity	0.45 (0.07)	0.317, 0.590
High functional diversity (+1 SD)	0.83 (0.10)	0.637, 1.02
Indirect effect of customer-oriented boundary spanning on customer adoption (H1b)		
	Effect (SE)	95% CI
Shared objectives and information elaboration	0.11 (0.04)	0.039, 0.208
Conditional indirect effects of customer-oriented boundary spanning on information elaboration through shared objectives (H2a)		
	Effect (SE)	95% CI
Low functional background diversity (−1 SD)	0.03 (0.04)	−0.038, 0.138
Mean functional background diversity	0.17 (0.09)	0.019, 0.356
High functional background diversity (+1 SD)	0.31 (0.14)	0.038, 0.604
Index of moderated mediation		
	Index (SE)	95% CI
Functional background diversity	0.14 (0.06)	0.017, 0.266
Conditional indirect effect of customer-oriented boundary spanning on customer adoption through shared objectives and information elaboration (H2b)		
	Effect (SE)	95% CI
Low functional background diversity (−1 SD)	0.01 (0.01)	−0.005, 0.026
Mean functional background diversity	0.03 (0.02)	0.001, 0.067
High functional background diversity (+1 SD)	0.05 (0.03)	0.002, 0.112
Index of moderated serial mediation		
	Index (SE)	95% CI
Functional background diversity	0.02 (0.01)	0.001, 0.048

Note: $N = 144$. Values are standardized coefficients (with standard errors).
Abbreviation: CI = confidence interval.

service variable had no significant effect on customer adoption ($\beta = 0.19$, $SE = 0.16$, $p = 0.227$).

Overall, the results of our analyses were qualitatively similar to the results excluding control variables, as reported in Table 4. We still found significant moderated serial mediation with customer adoption R^2 going down from 0.77 to 0.74 without controls. This shows that we “can rule out the controls as a potential explanation for the findings” (Becker 2005, 286).³

In sum, results for customer adoption supported predictions with a positive effect of customer-oriented boundary spanning as mediated by shared objectives and team information elaboration that was stronger for teams with higher functional background diversity. Results for information elaboration and the mediation hypotheses also supported predictions where information elaboration mediates the main effect of customer-oriented boundary

spanning and the interaction effect of shared objectives by functional background diversity clearly demonstrating that we cannot dismiss the role of information elaboration in the effects on customer adoption (cf. van Knippenberg and Hoefer 2021).

4 | Discussion

We argued that customer-oriented boundary spanning is an important influence on customer adoption by stimulating information elaboration with a focus on the customer's perspective. Drawing on information elaboration theory (van Knippenberg et al. 2004) and the attention-based view (Ocasio 2011) as well as research on professional service teams (Gardner et al. 2012), we argued this would be evident in both a main effect of customer-oriented boundary spanning and moderated by functional background diversity. These predictions were supported—and

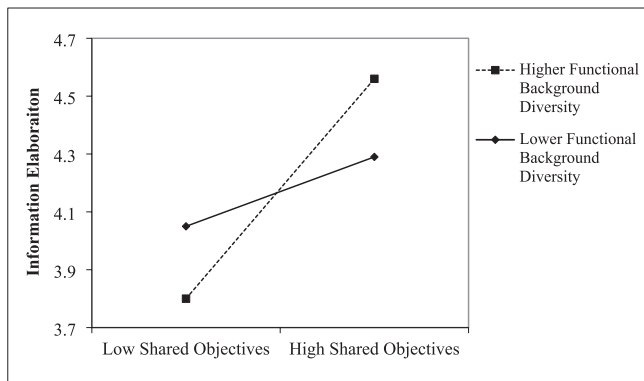


FIGURE 2 | Moderating effect of functional background diversity on information elaboration.

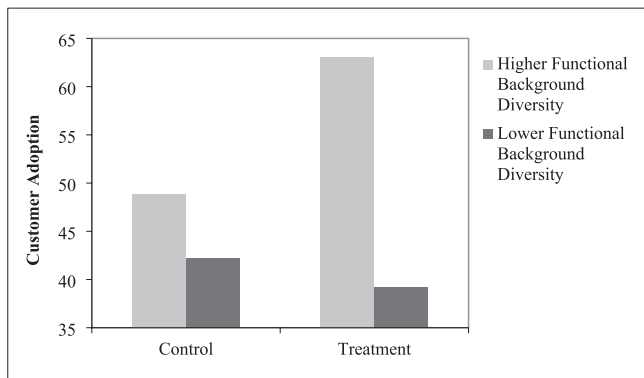


FIGURE 3 | Direct effect of customer-oriented boundary spanning on customer adoption moderated by functional background diversity.

importantly so with causal evidence for the role of customer-oriented boundary spanning and mediating evidence for the roles of shared objectives and information elaboration to support our theoretical analysis.

4.1 | Theoretical Implications

The current evidence for the positive effect of customer-oriented boundary spanning on customer adoption complements earlier work on boundary spanning and team information elaboration. This earlier work considered boundary spanning more broadly (e.g., Alexander and van Knippenberg 2014; Ancona and Caldwell 1992; van Knippenberg et al. 2024) rather than specifically focused on the external user (customer) of the team's output. Such boundary spanning is important in allowing the team to search for the information it needs and where it is most likely to find it. Leveraging information elaboration theory and the attention-based view, we theoretically explain that when the team is insufficiently focused on the customer (e.g., because the team's information search is guided primarily by members own functional perspectives), the team's information elaboration process risks being insufficiently attuned to the customer's needs. The extant adoption literature has long recognized that firms should better understand their customers' unmet needs to foster adoption (Karp 2023; Frambach and Schillewaert 2002). Our study provides a meaningful extension of the customer success and adoption literature as it shows that customer-oriented boundary spanning addresses core challenges in adoption, and it has a significant effect as it helps mobilize team informational resources in the service of customer adoption. While boundary spanning that is not limited to the customer may be associated with a broader scope of information search, this may come at the

TABLE 4 | Effects of customer-oriented boundary spanning on customer adoption, information elaboration, and shared objectives without control variables.

	Shared objectives	Information elaboration		Customer adoption	
		Model 1	Model 2	Model 1	Model 2
Constant	−0.19 (0.12)	−0.18 (0.10)	−0.13 (0.08)	−0.11 (0.07)	−0.14 (0.07)
Customer-oriented boundary spanning	0.36 (0.16)*	0.33 (0.15)*	0.26 (0.12)**	0.20 (0.10)*	0.14 (0.10)
Shared objectives	—	0.25 (0.07)**	0.44 (0.06)**	0.06 (0.05)	0.02 (0.05)
Functional background diversity	—	0.44 (0.07)**	0.16 (0.07)*	0.19 (0.05)**	0.10 (0.06)
Functional background diversity × shared objectives	—	—	0.38 (0.04)**	—	0.05 (0.04)
Functional background diversity × customer-oriented boundary spanning	—	—	—	—	0.32 (0.13)*
Information elaboration	—	—	—	0.13 (0.06)*	0.17 (0.06)**
Adoption baseline	−0.15 (0.08)	−0.17 (0.07)*	−0.10 (0.06)	0.84 (0.05)**	0.85 (0.05)**
R ²	0.05*	0.39**	0.60**	0.73**	0.74**
F	4.03*	21.85**	42.09**	74.21**	65.07**

Notes: N = 144. Values are standardized coefficients (with standard errors).

* $p < 0.05$.

** $p < 0.01$.

price of the focus on the customer—and thus be less effective in producing outcomes that engender customer adoption. From this perspective, the present analysis is not intended to be positioned in opposition to boundary spanning more broadly, but to complement earlier work with a specific focus on customer adoption. We considered customer-oriented boundary spanning as particularly effective in stimulating the shared objective of pursuing customer adoption (by putting the customer's perspective center-stage) and direct the team's attention to prioritize it accordingly.

From this perspective, a valuable development of the current analysis in future research would consider more explicitly—conceptually and empirically—how customer-oriented boundary spanning and boundary spanning targeting other sources of information may complement and influence each other (e.g., customer information may suggest the need for information from other sources external to the team). The development of this analysis would be particularly valuable in advancing information elaboration theory, professional service teams research, and customer adoption.

We contend that the present findings also have important theoretical implications for information elaboration theory and research beyond the role of customer-oriented boundary spanning specifically. Customer-oriented boundary spanning is an important focus for customer-facing teams, but not all teams are customer-facing. At a higher level of abstraction, the conclusion following from our findings is that an influence that strengthens shared objectives stimulates teams to make better use of their informational resources in information elaboration, and thus to perform better (cf. evidence that shared objectives are a positive influence with greater diversity, van Knippenberg et al. 2011). This is a conclusion that should also apply to teams that are not customer-facing. The current findings do not speak to this but underscore that exploring this possibility in future research would be highly worthwhile. The overarching idea would be that influences that strengthen the focus on teams' intended objectives (e.g., team leadership advocating a specific focus; van Knippenberg et al. 2013) can stimulate effective use of informational resources and thus have positive effects on team knowledge work outcomes both as a main effect and in interaction with factors that reflect the quality of teams' informational resources, such as functional background diversity.

The attention-based view is associated more with the organizational level of analysis and associated organization theory and strategic management perspectives (Ocasio 1997, 2011) than with the organizational behavior perspective of information elaboration theory (De Dreu et al. 2008; van Knippenberg et al. 2004). Viewed from that angle, the current research also contributes to the attention-based view by underscoring the value of considering the micro-foundations of the attention-based view in considering how attention expresses itself in knowledge integration at the level of the behavior of individuals and teams (cf. Weiser et al. 2020). From the perspective that information elaboration theory and the attention-based view complement each other at the team level of analysis (as per the current study), this gives rise to the question of whether the attention-based view will benefit from integrating insights

from information elaboration theory (e.g., concerning diversity of informational resources, motivational influences, etc.) in considering knowledge integration at the organizational level of analysis.

4.2 | Implications for Practice

Companies like the one that provided our study opportunity have a strong business interest in seeing their teams' tailored business-to-business solutions being strongly adopted by team clients as customer adoption of innovative solutions is a major indication of successful performance in this industry. From that perspective, the customer-oriented boundary spanning intervention is particularly attractive. The current study provides causal evidence for its effectiveness both in and of itself and in stimulating the team to make better use of its functional diversity. The current study also shows how it can be implemented in practice by the creation of a designated customer-oriented boundary spanning role based on a focused training for these role holders. At the same time, we also cautiously note that even though the intervention worked for the company we studied does not guarantee it will work elsewhere. However, adopting the intervention is a straightforward and a very actionable implication of the current study.

Customer-oriented boundary spanning practices in the IT industry, generally speaking, help to break down silos and reduce the risk of being “too creative,” in the sense of solely focusing on idea generation without or little thought about planning, prioritization, feasibility, resource mobilization, and change management. The integration of information across silos helps teams drive business and technology alignment necessary for sustainable adoption success.

The current findings also add to the case to compose functionally diverse teams. There is meta-analytical evidence that functional background diversity can be a positive influence on team innovation and performance, but this evidence also points to the need to take the conditions that foster this positive effect into account (van Knippenberg et al. 2020). The present findings are well-aligned with this conclusion and show that customer-oriented boundary spanning as an actionable intervention allows organizations to get more return on investment for efforts to compose functionally diverse teams. Put differently, the current findings suggest that efforts to create functionally diverse customer engagement teams are better spent when combined with a customer-oriented boundary spanning intervention than stand-alone.

4.3 | Limitations and Future Directions

The current study has clear strengths in its field experimental design, which allows for causal conclusions about the role of customer-oriented boundary spanning, and its reliance on objective client usage data to assess customer adoption. Like any study, however, it is not without limitations. A first and obvious limitation is about generalizability of the findings of data collected within a single enterprise software organization.

We note that we do not have data about the specifics of the solutions that were developed. As a result, we do not know how exactly customer-oriented boundary spanning influenced the solutions the teams developed for their clients. Our theory suggests that greater adoption as a function of customer-oriented boundary spanning results from the solutions being better aligned with customer needs. We do not have direct evidence to speak to this however, or to speak to whether solutions may also vary on other dimensions. Future research that would speak to these issues would be important in further developing the current analysis of increased focus on the customer. In that vein, we note that the shared objectives measure (Anderson and West 1998; van Knippenberg et al. 2011) has a long history in team research (Hülsheger et al. 2009) and captures the extent to which teams agree on objectives, but it does not capture the content of the shared objectives. The positive effect of customer-oriented boundary spanning we observed can be plausibly interpreted as resulting from a stronger focus on the customer's perspective, but it is a limitation that we do not have data that directly shows the higher shared objectives reflected a stronger focus on the customer. Similarly, we used the empirically validated measurement items from Faraj and Yan (2009) to measure the boundary spanning treatment check, which are not specific to customer-oriented boundary spanning. In doing so, we work on the assumption that an experimental effect of customer-oriented boundary spanning on this measure can be plausibly interpreted as reflecting higher customer-oriented boundary spanning specifically—but this assumption does not equate evidence and thus be seen as a limitation of our data.

Functional background diversity is a variable that can be relatively objective assessed, but it was not under experimental control other than through the fact that we randomly assigned teams to conditions. As a result, there were no confounds between customer-oriented boundary spanning and functional background diversity by design. Because diversity was not under experimental control, we do not know what it may have covaried with that might also influence results. Whereas this is consistent with field research in team diversity, it does mean that for stronger evidence there would be value in a future field-experimental opportunity to bring team functional background diversity under experimental control for causal conclusions.

5 | Conclusion

Customer adoption of the team deliverables is core to the success of knowledge teams focused on creating solutions for customers. This makes a focus on the user's perspective important for team's success in seeing the results of their efforts adopted by this external customer. From this perspective, we argued and found that customer-oriented boundary spanning is a useful and actionable way to increase customer adoption. The current evidence in support of our analysis clearly sets an agenda to further develop this analysis and extend information elaboration theory.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

¹ When we excluded all 30 multi-team service cases and analyzed the remaining sample of 114 team cases, we still found significant serial moderated mediation and all hypotheses hold. Furthermore, we regressed customer adoption on multi-team service and internal assignment indicators along with our other study variables and found that both indicators had no significant effect in the treatment group (comparing these two arrangements to the ideal design). This shows that we can rule out these deviations from the ideal design as a potential explanation for our finding.

² The concept and measure have been referred to as clarity of objectives and as shared objectives by the original authors of the conceptualization and operationalization. It captures well what we focus on—shared objectives stimulating shared attention in information elaboration.

³ We report our results in tables with standardized coefficients. Tables 2–4 with unstandardized coefficients are available as supporting information in the online appendix of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.